

EEE 419/591

Summ 2026

Homework 2

[Same instructions as HW 1. Changes highlighted in bold font]

Solve the following problems. Put them into a single code named **hw2.py** and submit it to the designated link on Canvas. If you upload a second attempt to Canvas and it renames your file, ignore the renaming.

In this HW, only limited AI is allowed. Examples of allowed prompts on this HW:

- What is a greatest common divisor of two numbers?
- How do I write an if statement in python? Give me an example
- How do I check if a number is integer in python?
- I got this error message. How do I interpret it?

Examples of prohibited prompts on this HW:

- Write me a code that calculates the greatest common divisor of a list of numbers in python
- I ran this code. [PASTE CODE HERE]. I got this error message. [PASTE THE ERROR MESSAGE HERE]. How do I fix my code?

Do not forget to cite your AI chat if you used any. Check the Syllabus document for details.

Problem 1: Estimating the accuracy of quad Package for Finite Integrals

Compute the value of the integral:

$$I_1(r) = \int_{-4}^5 (ax^5 + bx^2 - c)dx$$

where a , b , and c are some numbers fed by the user and ranging between -10 and 10. Pay attention to the powers of x in the integrand. Pay attention to the powers of x . The integrand is not a quadratic function. Use the following two methods to compute the above integral

Method 1: Use `scipy.integrate.quad` package.

Method 2: Numerical integration with a suitable step size. Recall: Numerical integration calculates the area under the curve manually.

Your code should expect a set of 3 numbers separated by a space as an input and print the value of the integration for each method as well as the percentage error between the two methods as the following example [Numbers shown might not reflect the above problem]:

Input a set of 3 numbers between -10 and 10: -2 6 0

Method 1: I1 = -8.3333

Method 2: I1 = -8.3333

Percentage error: 0.0005%

Hint: Choose a small step size in Method 2 to be considered more accurate. It is up to you to decide how small is “small”. You need to balance the accuracy with running time. This is an important skill you learn. Choosing a very small step size for ultra-high accuracy might not be right/wise. Note: it is acceptable to get 0 error in this problem since quad is already accurate.

$$\text{Percentage error} = \left| \frac{\text{Accurate Method} - \text{Inaccurate Method}}{\text{Accurate Method}} \right| \times 100\%$$

Problem 2: Estimating the Accuracy of the Substitution Method and Numerical Method for Infinite Integrals

Compute the value of the integral:

$$I_2(r) = \int_0^{\infty} \frac{dx}{1+x^2} = \frac{\pi}{2}$$

using the following two methods:

Method 1: Substitution method taught in video lecture 2.6 at min ~3:45. The substitution itself can be done via code or on a piece of paper, but the integration must be computed via the integration command in python. Your code should clearly show the new integration limits that result from the substitution method. You might be able to find a package that will integrate this integral without substitution, do not use it.

Method 2: Numerical Integration: use a suitable step size and value to replace ∞ . Recall: Numerical integration calculates the area under the curve manually. Instead of Numerical Integration, you are allowed to use a package that calculates infinite integrals without substitution.

Note that this is a known integral whose expected answer is, in fact, $\pi/2$. For each method,

- Print out the integration value to a precision of 8 digits past the decimal point.
- Print out the absolute difference from the expected answer. Recall, you already have an accurate value of π in the numpy package. Subtract the expected answer from the integration value from each method. Print as many decimal digits as what illustrates the difference (not more than 16 decimal digits).

Your code should not expect anything as an input and prints something like the following example (on each line, print the number first then the name of the method for easy comparisons):

```
3.14159265          # I2 Method 1
3.14159245          # I2 Method 2
0.0000000000000099  # Difference Method 1
0.0000000000000043  # Difference Method 2
```

Note that these numbers might be inaccurate. Your numbers will be different.

MANDATORY – Reminder to cite AI chat